SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

ESS PCS 100kW (ESS-PCS-100kW) — Datasheet

1. Overview

Bidirectional power conversion system 100kW for battery energy storage. Product family: Energy storage systems. Lifecycle status: **ACTIVE**.

2. Technical specifications

2.1. Ratings

Item	Value
Rated AC power	100 kW
AC grid connection	3-phase 400 V, 50/60 Hz
DC voltage range	600-900 V

2.2. Performance and characteristics

Item	Value
Max efficiency	97.5%

2.3. Endurance and communication

Item	Value
Battery interface	CAN 2.0B to BMS
SCADA interface	Modbus TCP

2.4. Protection and options

Item	Value
Anti-islanding	IEC 62116 compliant
Grid code profiles	Selectable country profiles (parameter G-01)

3. Installation data

3.1. Terminals

- Dimensions (W x H x D): 1000 x 800 x 2000 mm
- Weight: 520 kg
- Degree of protection: IP54
- Operating temperature: -20 to +50 °C (derating above 45 °C)

4. Standards and certifications

Applicable standards:

- IEC 62477-1
- IEC 62116
- IEC 61000-6-2/-4

Certifications held:

- CE
- KC
- UL 1741 SB (-U variant)

Certificates and declarations of conformity are available from the Certification & Compliance team on request. Certifications not listed above are not held for this product; project-specific evaluation can be requested.

5. Ordering information

Item	Value
Ordering code	ESS-PCS-100kW
Standard lead time	10 weeks ex-works
Minimum order quantity	1
Warranty	36 months (see POL-WARRANTY)
Firmware	latest 2.70 (see RN-ESS-PCS-100kW)

6. Related documents

- IG-ESS-PCS-100kW — Installation and Commissioning Guide
- TG-ESS-PCS-100kW — Troubleshooting and Error Code Guide
- RN-ESS-PCS-100kW — Firmware Release Notes
- FAQ-ESS — Energy storage systems FAQ

SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

ESS PCS 100kW (ESS-PCS-100kW) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: 1,000 mm in front for maintenance; ambient controlled by HVAC between 15 and 30 °C
- Operating temperature: -20 to +50 °C (derating above 45 °C)
- Degree of protection: IP54
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Check battery rack insulation resistance
2. Verify CAN communication between PCS and BMS
3. Select grid code profile (parameter G-01)
4. Perform pre-charge test
5. Run charge/discharge test at 20% power

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
Grid over-voltage trip	Wrong country grid profile selected	TG-ESS-PCS-100kW §3.1
Frequency trip during generator operation	Firmware defect in ROCOF measurement	TG-ESS-PCS-100kW §3.2
IGBT over-temperature	Air filter clogged	TG-ESS-PCS-100kW §3.3
DC insulation fault	Moisture in battery rack DC cabling	TG-ESS-PCS-100kW §3.4

On-site commissioning support can be requested through the Global CS Center. For energy storage systems the request is handled by the technical support team.

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ESS PCS 100kW (ESS-PCS-100kW) — Firmware Release Notes

1. Summary

Latest firmware: **2.70**. Updating to the latest version is recommended for all units.

2. Version history

2.1. Version 2.70 (2026-03-30)

- Added: frequency-watt response curve for new grid codes
- Fixed: G.02 nuisance trip on ROCOF measurement during genset operation

2.2. Version 2.62 (2025-07-07)

- Fixed: G.06 BMS communication loss with BMS FW 3.4 heartbeat change

2.3. Version 2.50 (2024-11-11)

- Baseline release

3. Update procedure

1. Back up parameters / project before updating.
2. Connect the engineering tool (v4.6 or later).
3. Do not switch off power during the update.
4. After the update, verify the version and restore parameters if required.
5. If the update fails, do not power-cycle the unit repeatedly; contact support for the recovery procedure.

4. Compatibility

PCS firmware 2.62 or later is required with BMS firmware 3.40 or later (CAN heartbeat change).